

COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No: 06

Sequential Circuit



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Recap of Previous Lecture



- Asynch. Counter
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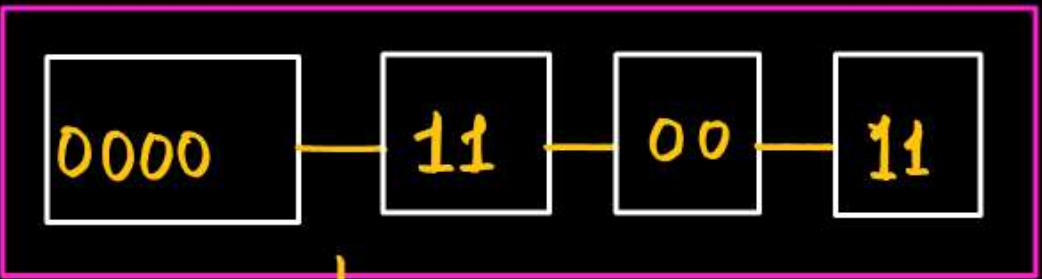
Topics to be Covered

Question Discussion

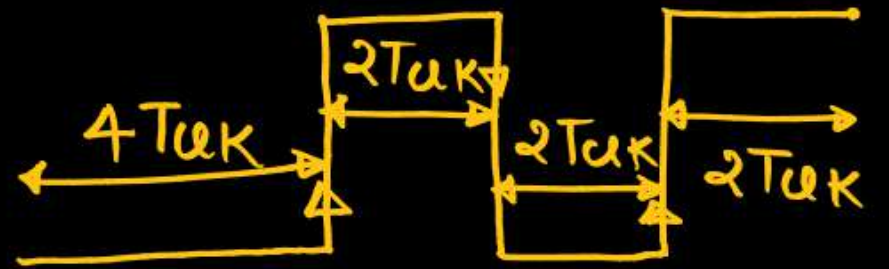
Max. frequency of operation.



$Q_1 \rightarrow$



$T_{Q_1} = 10T_{clk}$
 $f_{Q_1} = f_{clk}/10$



disjoint square waveform.

Q_3	Q_2	Q_1	Q_0
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0
1	0	0	1
1	0	1	0

MOD no. = 10

$Q_1 \rightarrow T_{ON} = 1T_{clk}, T_{OFF} = 1T_{clk}$

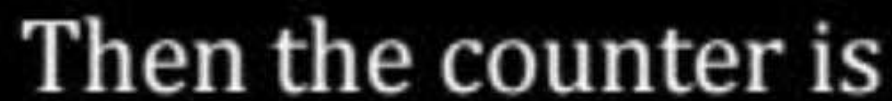
$T_{Q_1} = 2T_{clk} \rightarrow f_{Q_1} = f_{clk}/2$

$Q_2 \rightarrow T_{OFF} = 6T_{clk}, T_{ON} = 4T_{clk}$

$T_{Q_2} = 10T_{clk} \rightarrow f_{Q_2} = \frac{f_{clk}}{10}$

$Q_3 \rightarrow T_{OFF} = 8T_{clk}, T_{ON} = 2T_{clk}$

$T_{Q_3} = 10T_{clk}, f_{Q_3} = f_{clk}/10$

$$Q_z = 1$$
$$Q_x = 1$$


- (a) MOD-6 up counter (b) MOD-5 down counter
- (c) MOD-5 up counter (d) MOD-6 down counter

	Q_z	Q_y	Q_x
	0	0	0
↑	0	0	1
↑	0	1	0
↑	0	1	1
↑	1	0	0
↑	1	0	1

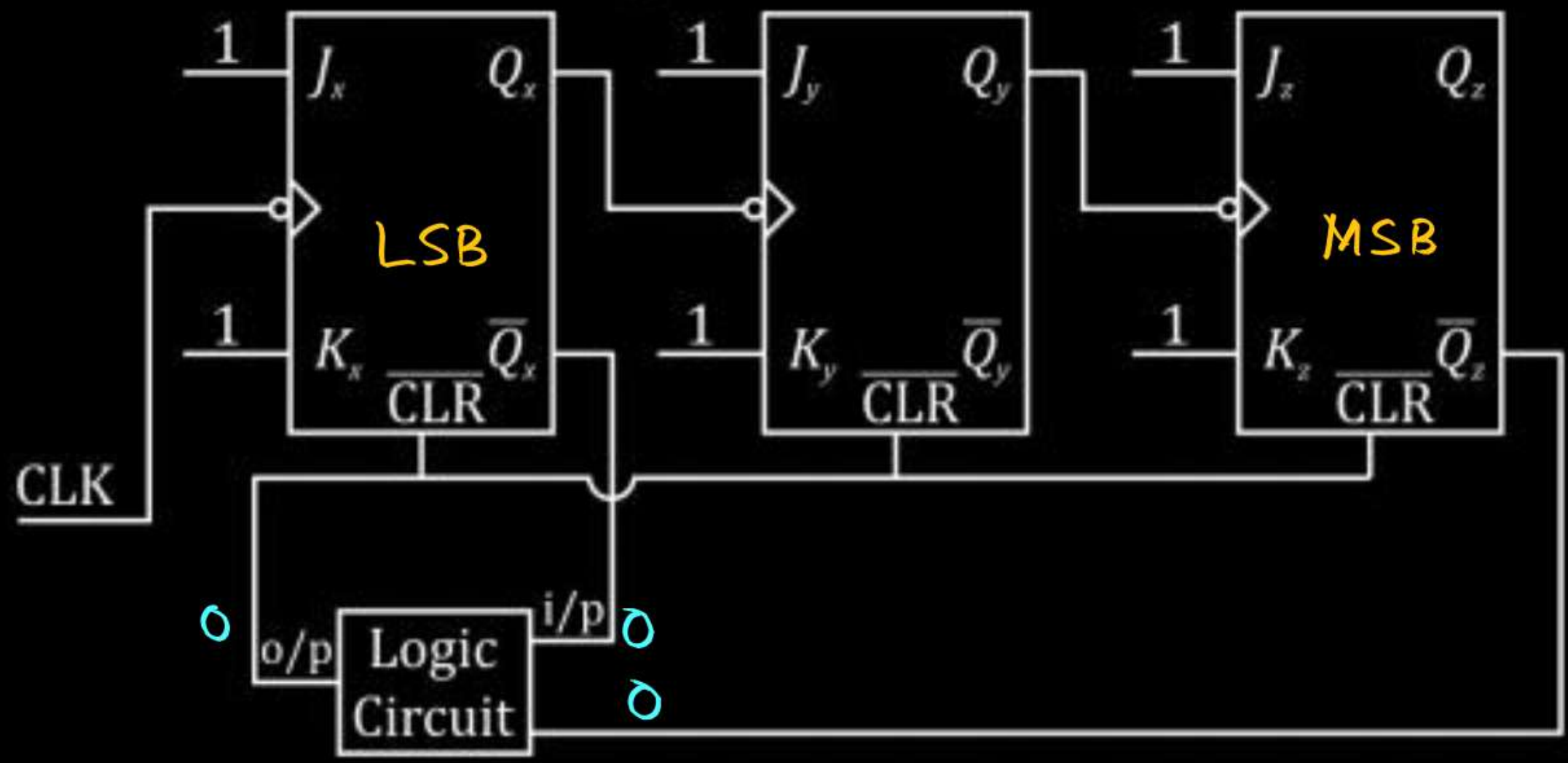
MOD-5

exist momentarily
and
not counted
in MOD no.

[Question]

up-counter

A sequential circuit is designed as shown :



Q_z	Q_y	Q_x
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1

For above circuit to work as MOD-5 counter, the logic circuit will be

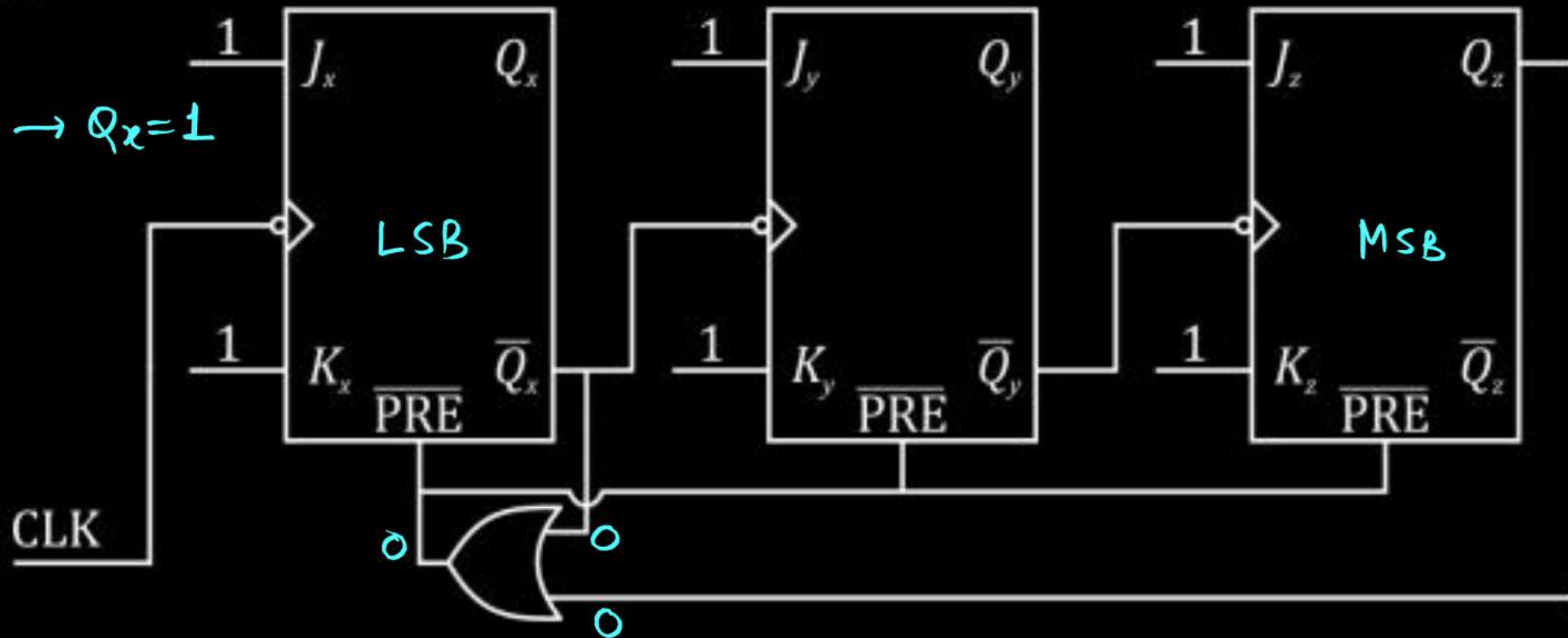
- (a) 2-input OR GATE
- (b) 2-input AND GATE
- (c) 2-input NAND GATE
- (d) 2-input NOR GATE

[Question]

A sequential circuit is as given below :

down counter

$Q_2 = 0$
 $\overline{Q_x} = 0 \rightarrow Q_x = 1$



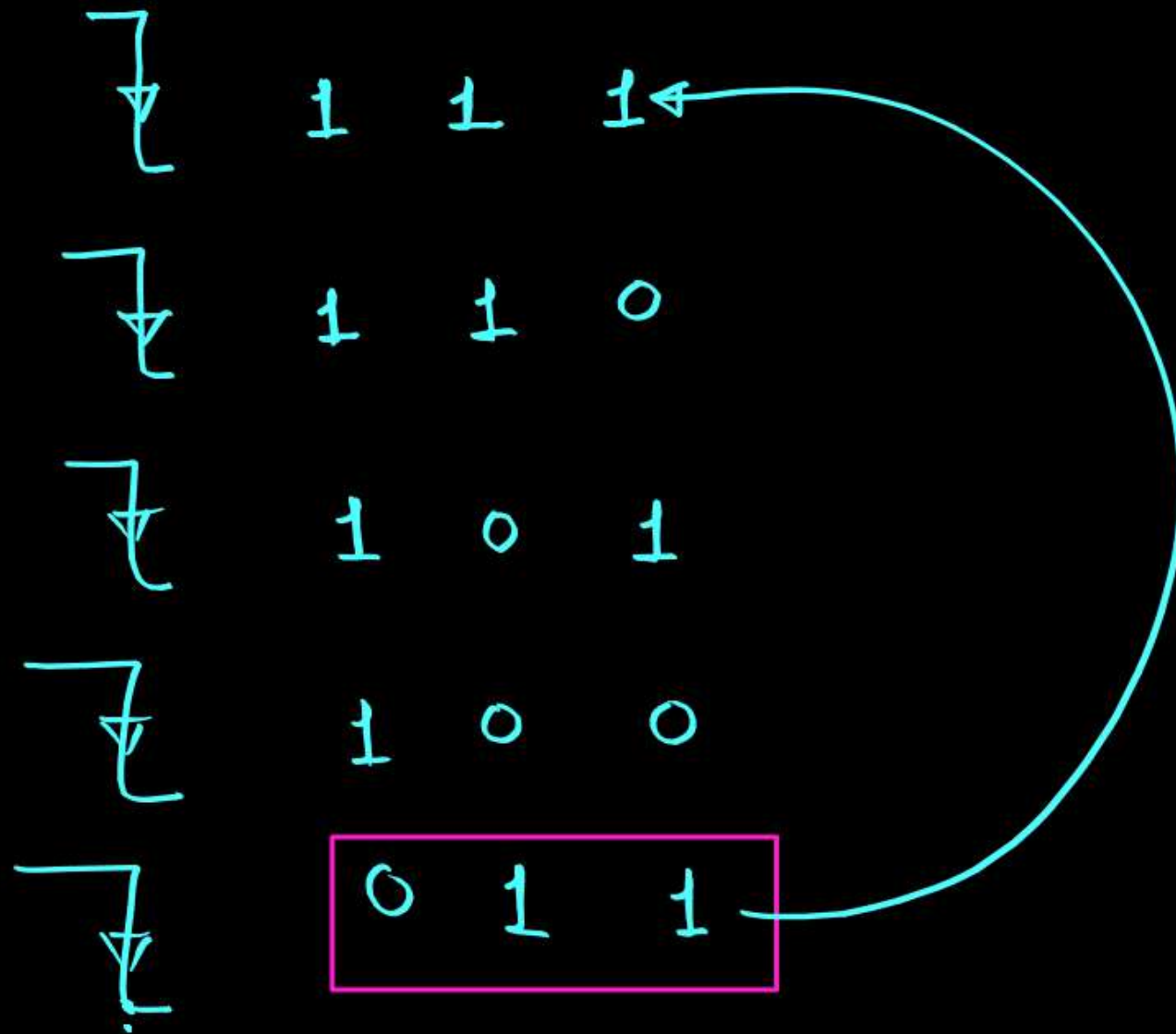
The circuit works as :

- (a) MOD-5 down counter
- (c) MOD-6 down counter

- (b) ✓ MOD-4 down counter
- (d) ✗ MOD-5 up counter

$Q_z Q_y Q_x$

0 0 0 → initial transient



MOD-4 down counter

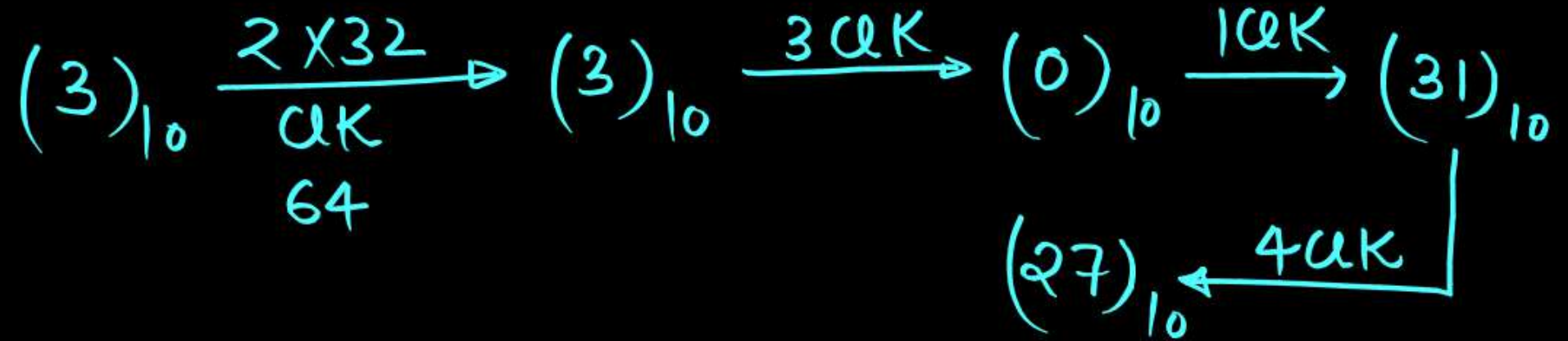
[Question]

We have a MOD-32 down counter. If its starting state is $(3)_{10}$ then after application of 72 clock pulses the count of the counter will be

$(27)_{10}$.

0-31

MOD no. = 32



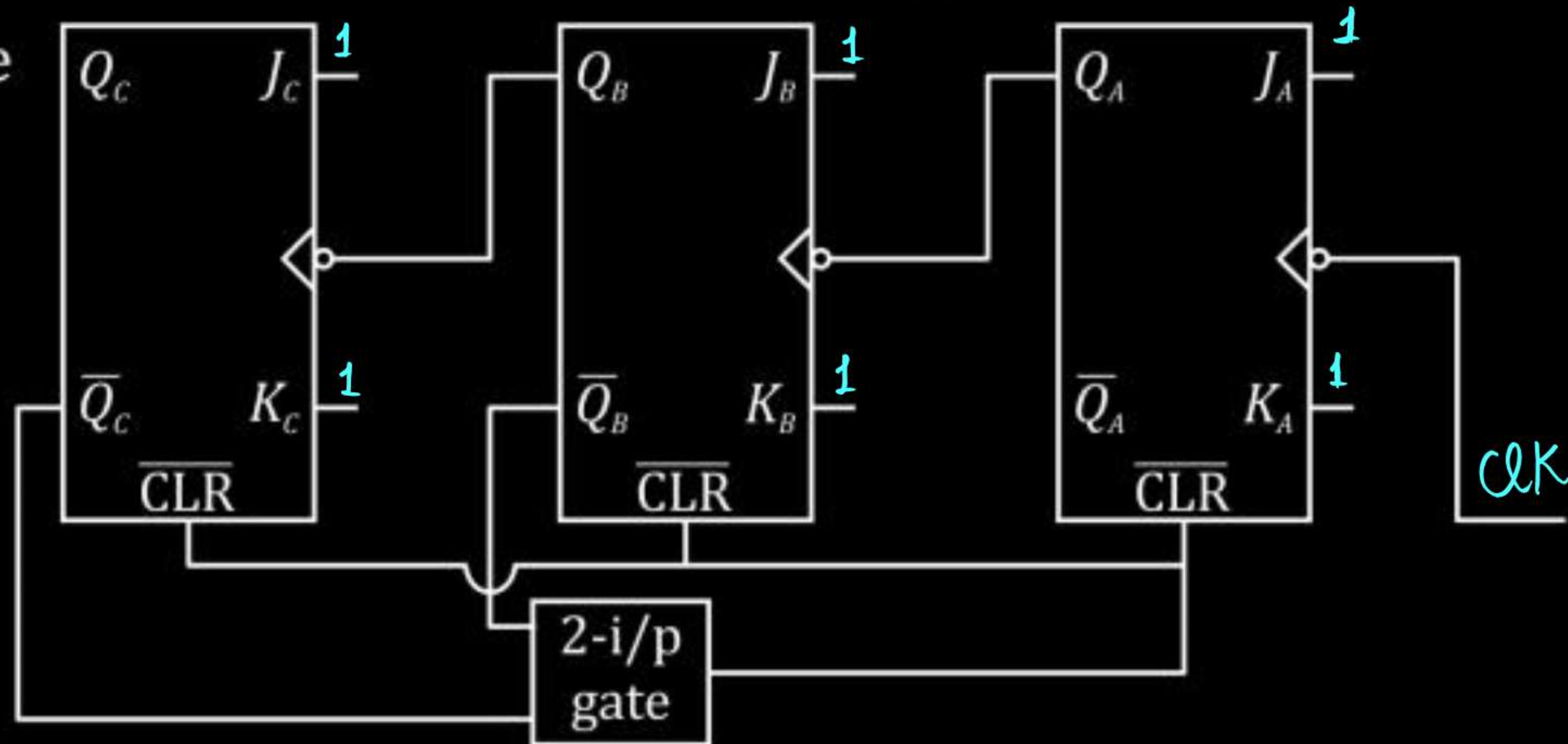
[Question]

In module-6 ripple counter shown in figure, the output of 2-input gate is used to clear the J-K FF.

H.W.

The gate will be

- (a) NAND
- (b) NOR
- (c) OR
- (d) AND

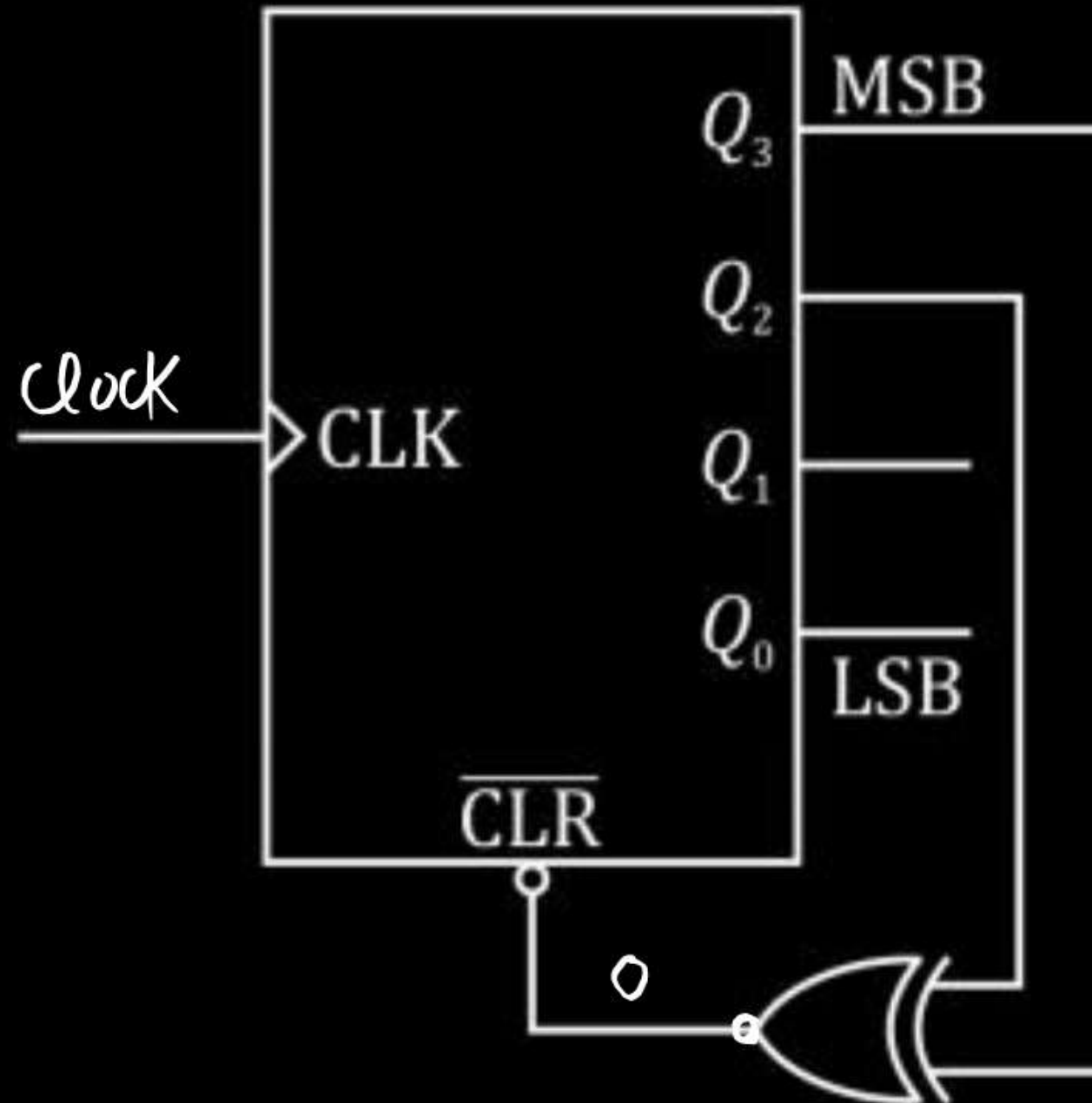


[Question]

The figure shows a binary counter with synchronous clear input with decoding logic shown. This counter works as

The counter is

- (a) MOD-2 counter
- (b) MOD-4 counter
- (c) MOD-5 counter
- (d) MOD-6 counter

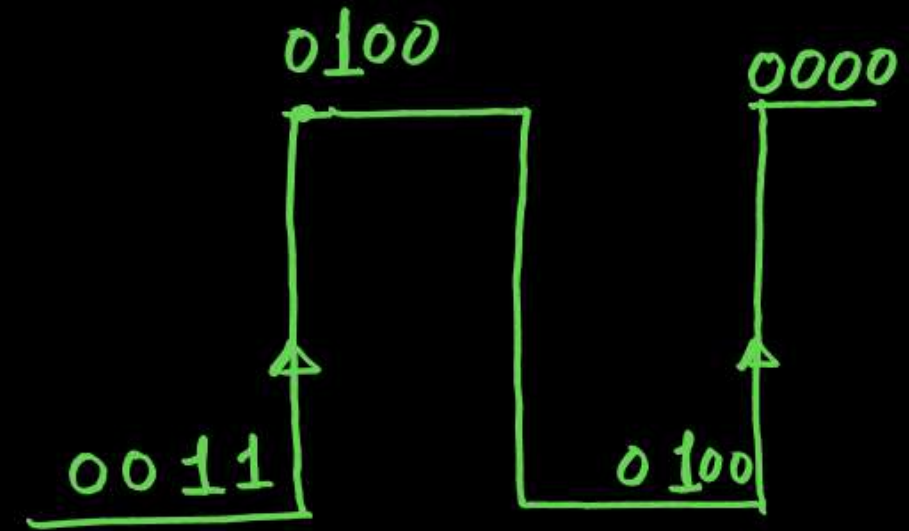




	Q_3	Q_2	Q_1	Q_0
	0	0	0	0
↑	0	0	0	1
↑	0	0	1	0
↑	0	0	1	1
↑	0	1	0	0

MOD-5

→ exist for one cycle
and counted in MOD no.





[Question]

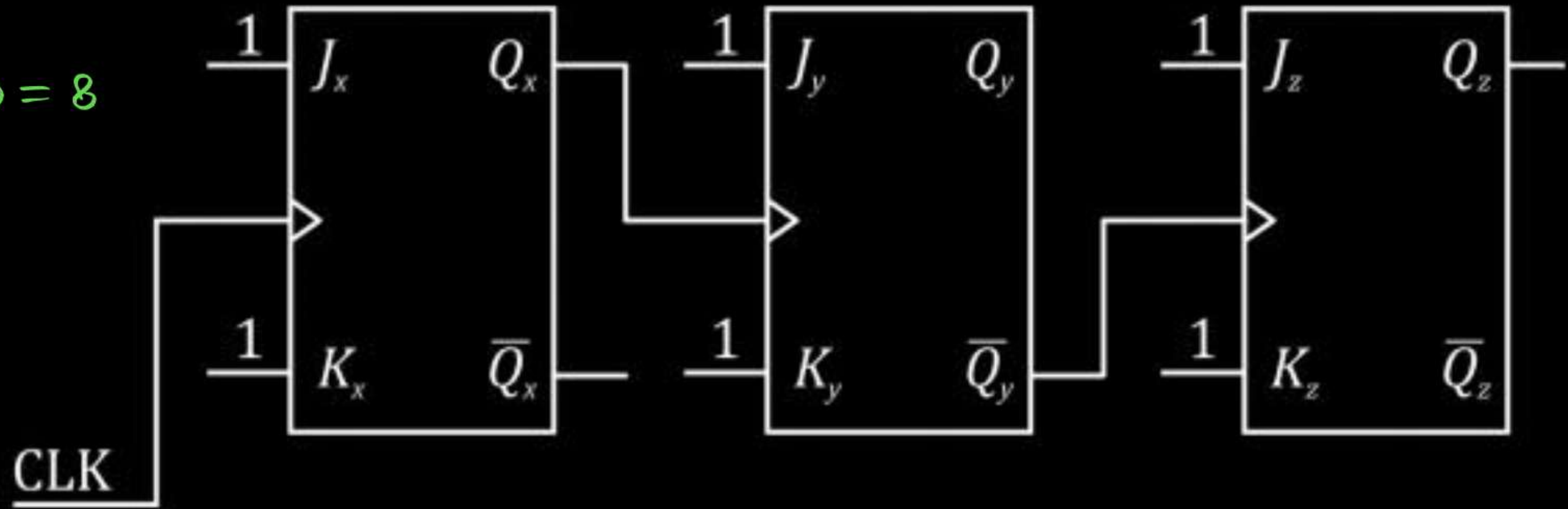
We have 10-bit ripple counter which is counting in up sequence. Two most significant bits are NAND together to generate an output y . Output y is periodic waveform with T_{ON} (time duration for which y is high in one period) $15000\ \mu\text{sec}$. Then the clock frequency (kHz) _____.

[Question]

A sequential circuit is as given below :

A.W.

MOD = 8

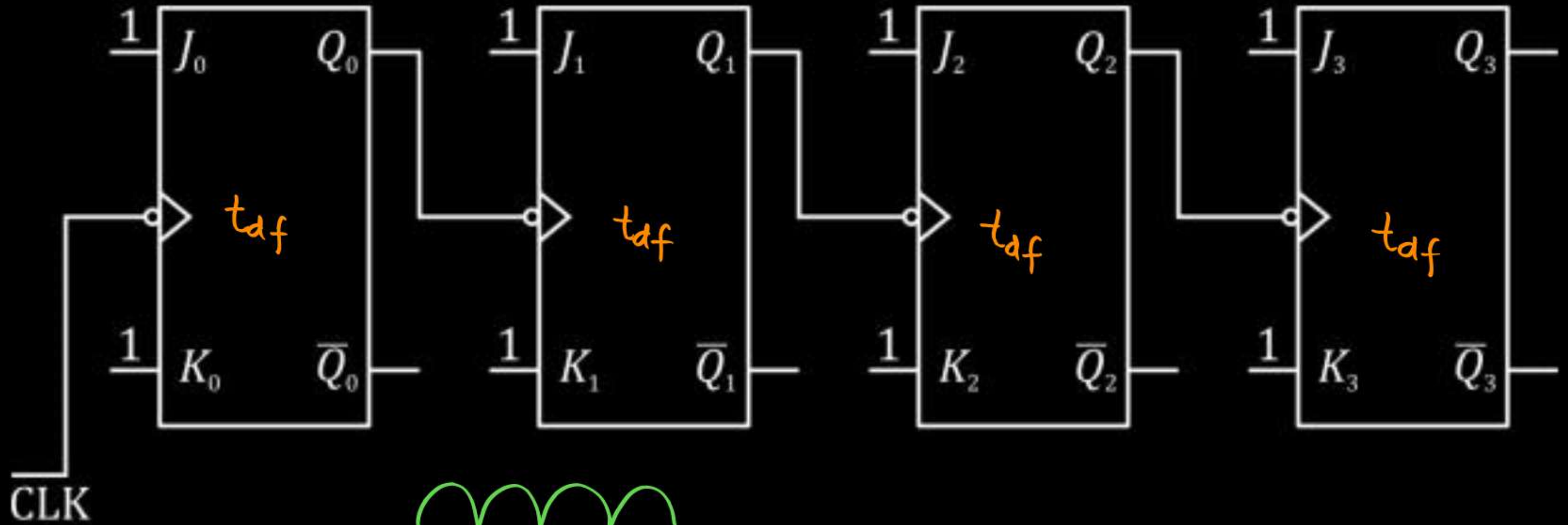


If starting state of the counter $Q_z Q_y Q_x$ is $(101)_2$ then after 251 CLK pulses output $Q_z Q_y Q_x$ is ()₁₀.

[Decoding Error and Max. Clock Frequency]

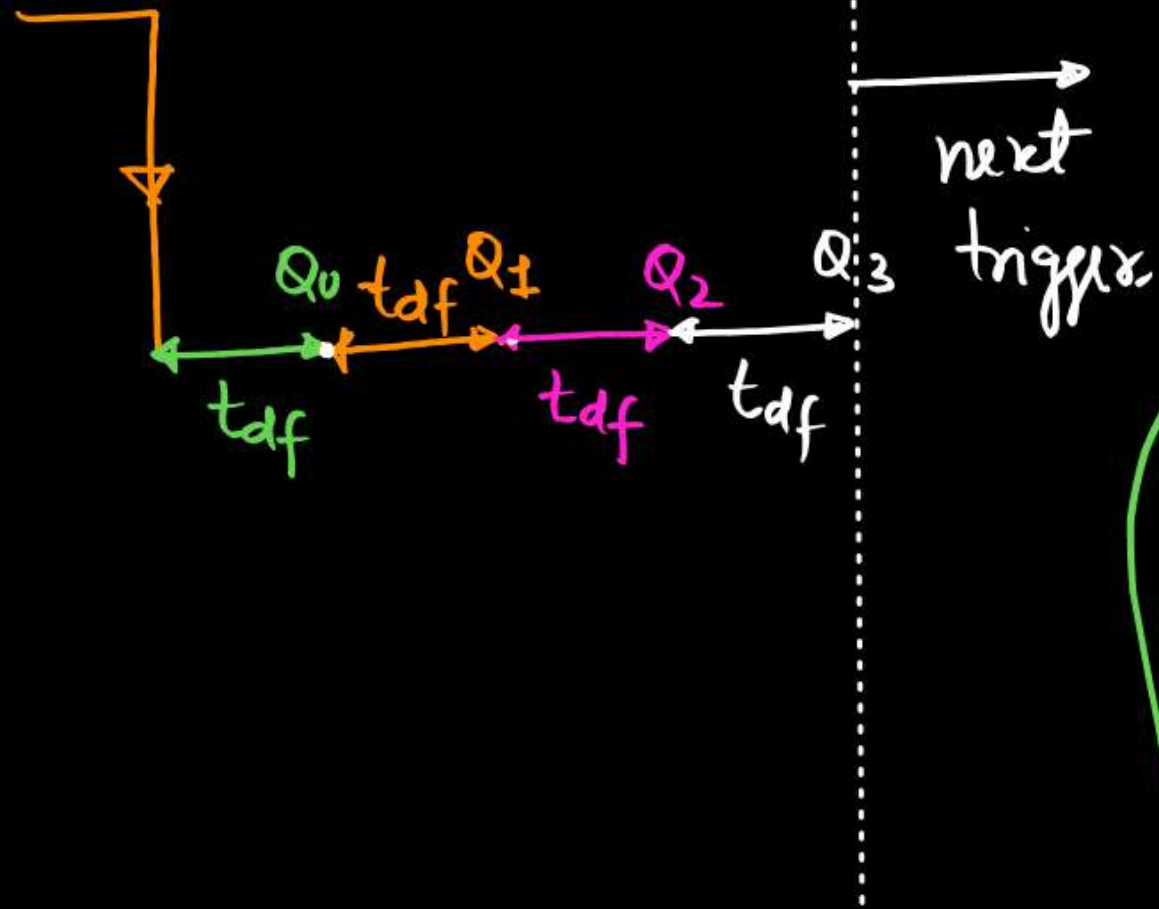


Lets take a 4-bit asynchronous counter :




Ripple counter

Lets understand the worst case delay and max clock frequency :



Worst case delay $T_d = n t_{df}$
for proper operation

$$T_{clk} \geq n t_{df} \rightarrow (T_{clk})_{min} = n t_{df}$$

$$\frac{1}{f_{clk}} \geq n t_{df} \rightarrow \frac{1}{n t_{df}} \geq f_{clk}$$

$$f_{clk} \leq \frac{1}{n t_{df}}$$

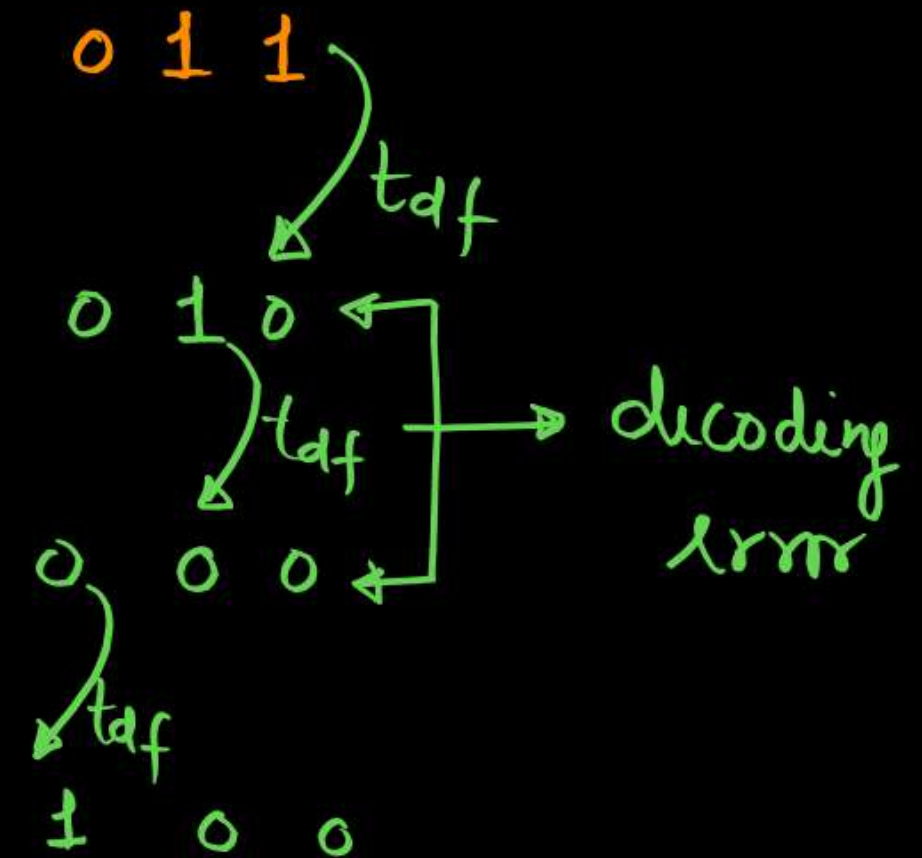
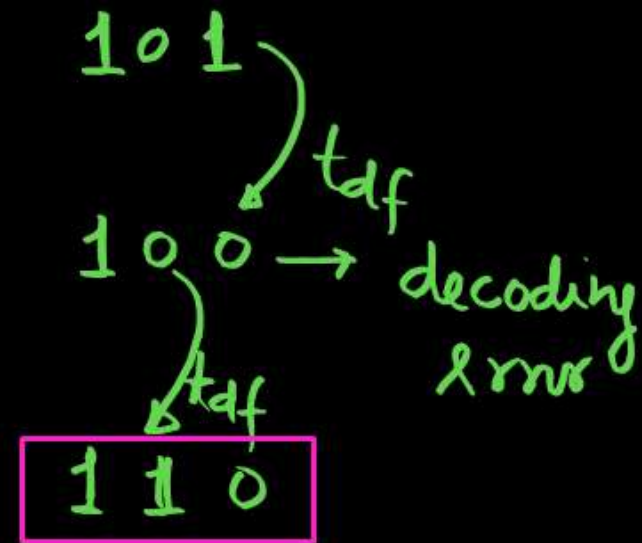
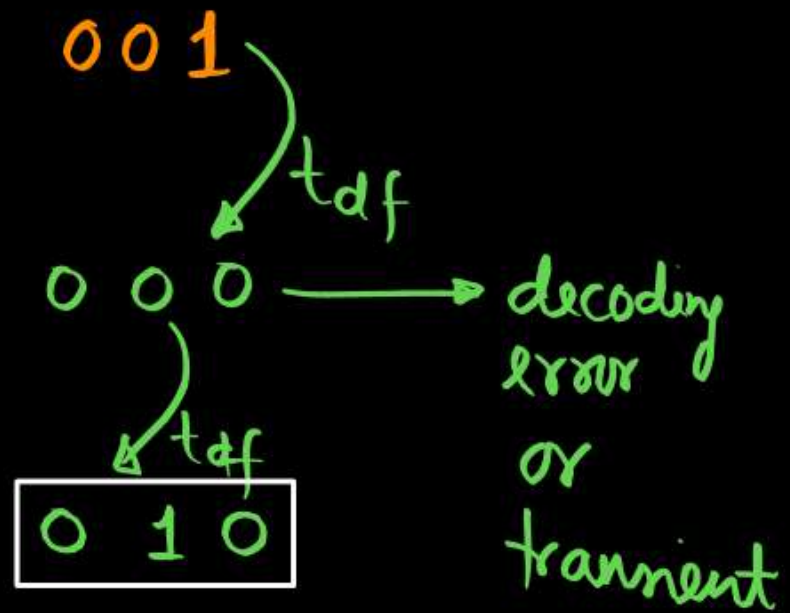
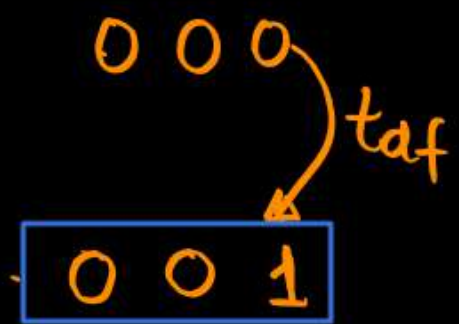
$$(f_{clk})_{max} = \frac{1}{n t_{df}}$$

$$\frac{1}{T_{clk}} \leq \frac{1}{n t_{df}}$$

$$f_{clk} \leq \frac{1}{n t_{df}}$$

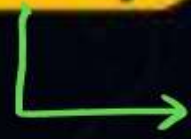
- IMP Points :

Decoding Error—:





Topic : 2 Min Summary



Asynchronous Counter.

Thank you

GW
Soldiers !

